ADVANCED DISTRIBUTED COMPUTING

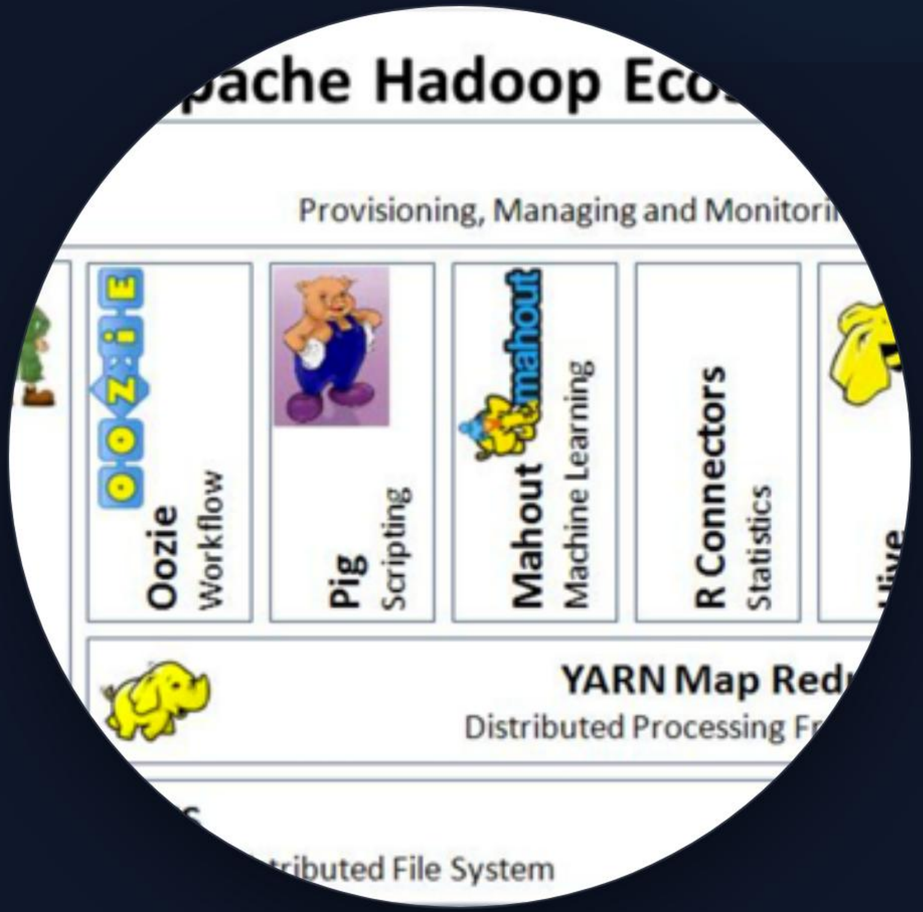
Mastering the Big Data Ecosystem: From Batch to Real-time

1. MAPREDUCE ARCHITECTURE

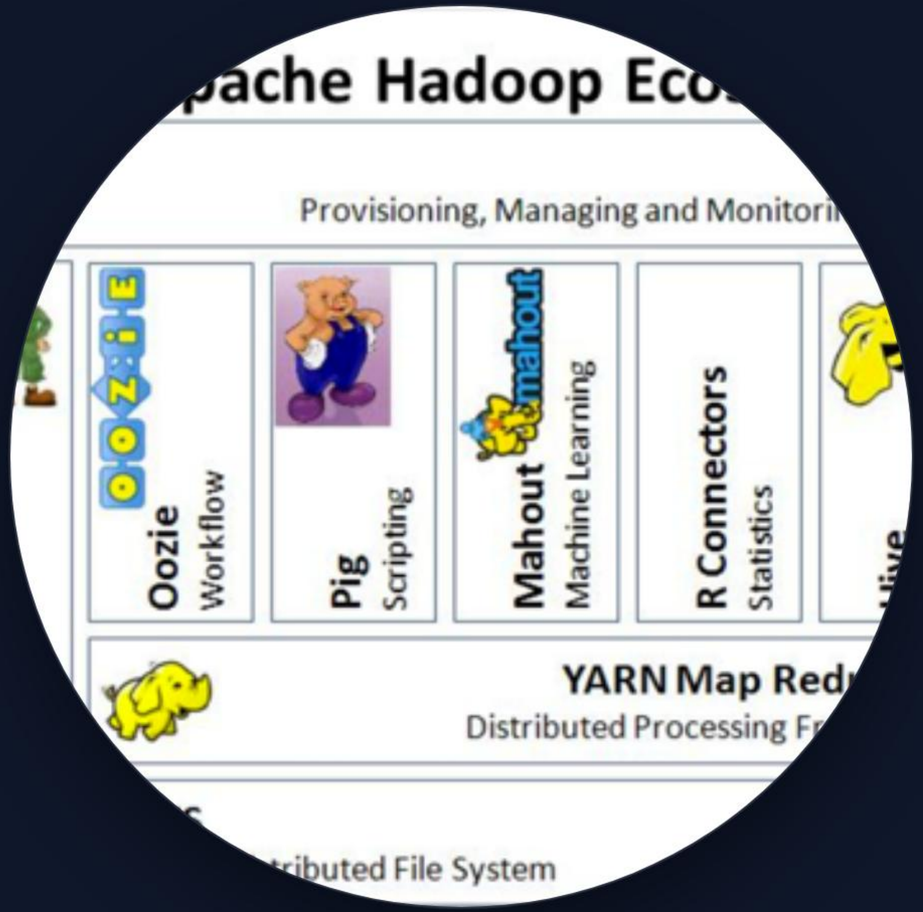
The Batch Pioneer

MapReduce is a programming model for processing large data sets with a **parallel, distributed algorithm** on a cluster.

- **Map:** Filters and sorts data (local processing).
- **Shuffle:** Transfers data to the right nodes.
- **Reduce:** Aggregates results for final output.



2. APACHE SPARK ENGINE



In-Memory Computing

Spark is a multi-language engine for executing data engineering, data science, and machine learning on **single-node machines or clusters**.

- **Speed:** Up to 100x faster than MapReduce via RAM.
- **Ease of Use:** High-level APIs in Java, Scala, Python, R.
- **Generality:** Supports SQL, Streaming, and ML.

3. SPARK VS. MAPREDUCE

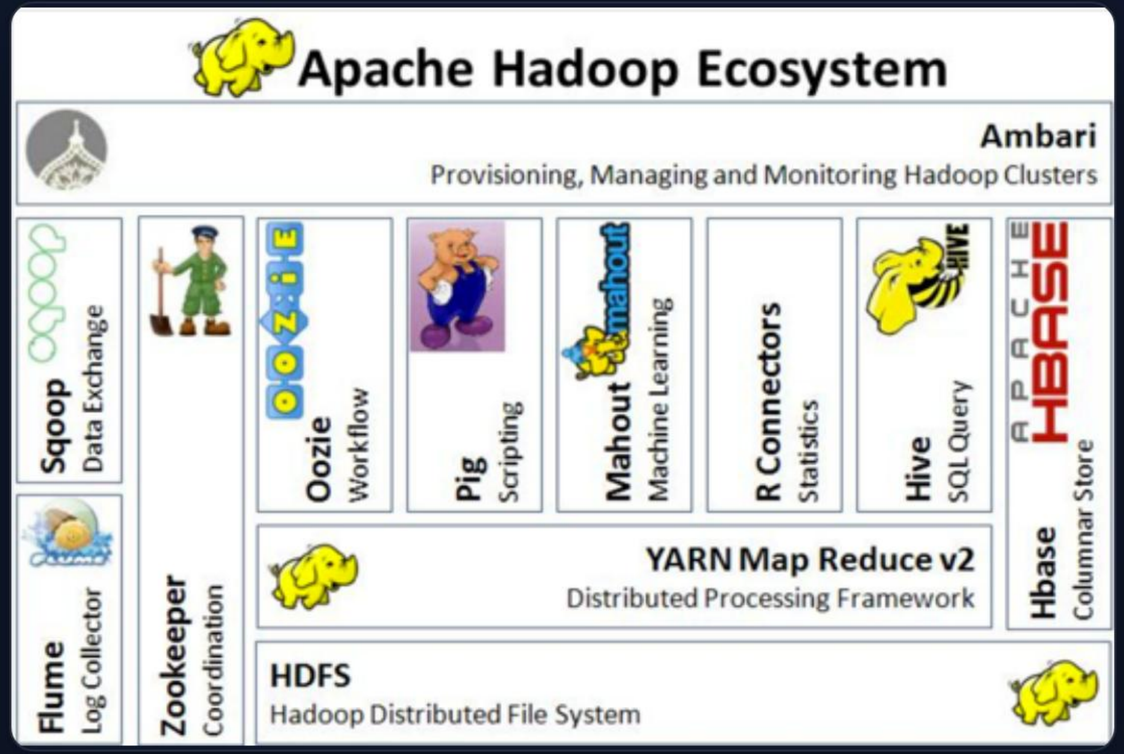
Feature	MapReduce	Apache Spark
Speed	Slow (Disk-bound)	Fast (In-memory)
Data Processing	Batch only	Batch, Real-time, Iterative
Fault Tolerance	Replication across HDFS	RDD (Resilient Distributed Datasets)
Ease of Use	Complex Java code	Rich APIs and SQL support

4. SPARK ON HADOOP CLUSTER

Spark is not a storage system; it relies on Hadoop's infrastructure for resource management and data persistence.

The Symbiosis

Spark runs on **YARN** (Yet Another Resource Negotiator) to manage cluster resources while reading/writing data from **HDFS**.



5. BIG DATA ECOSYSTEM TOOLS

Sqoop

SQL-to-Hadoop. Bulk data transfer between RDBMS and HDFS.

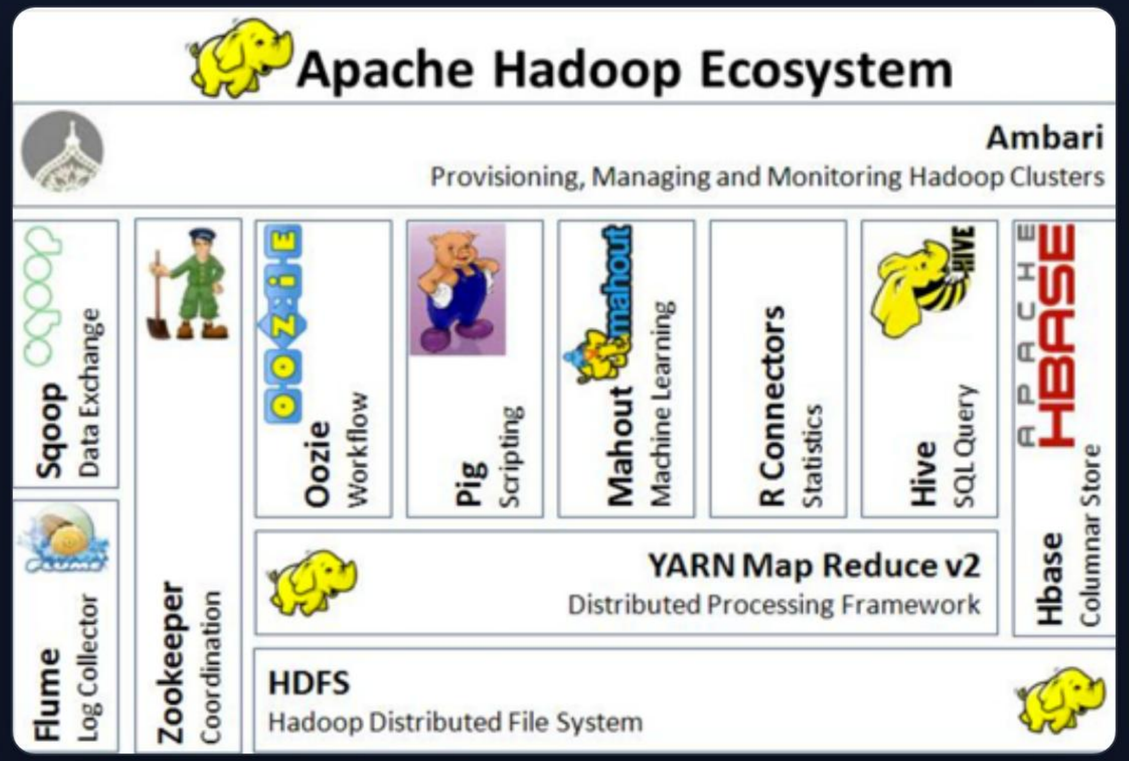
Flume

Distributed service for collecting and aggregating log data.

Oozie

Workflow scheduler system to manage Apache Hadoop jobs.

6. SQOOP: RDBMS TO HDFS



Structured Data Import

Sqoop uses MapReduce to import/export data, providing **parallelism** and **fault tolerance**.

```
sqoop import --connect
jdbc:mysql://db.host/corp
--table employees --target-dir
/user/hdfs/emp
```

7. FLUME: LOG COLLECTION

Source

Receives events from an external source like a web server.

Channel

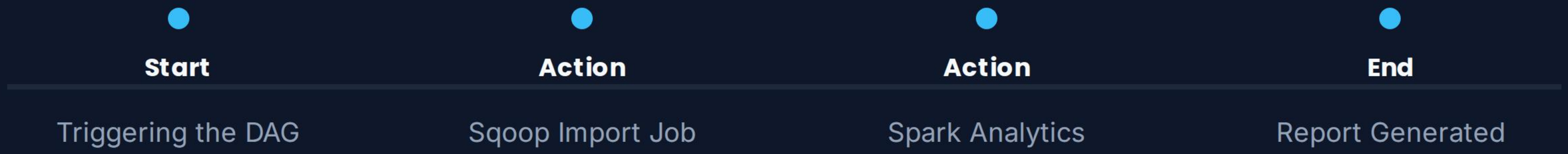
A passive store that keeps the event until consumed by a sink.

Sink

Removes the event from the channel and puts it into HDFS/HBase.

Flume is designed for **streaming log data** directly into the Hadoop core.

8. OOZIE: WORKFLOW SCHEDULER



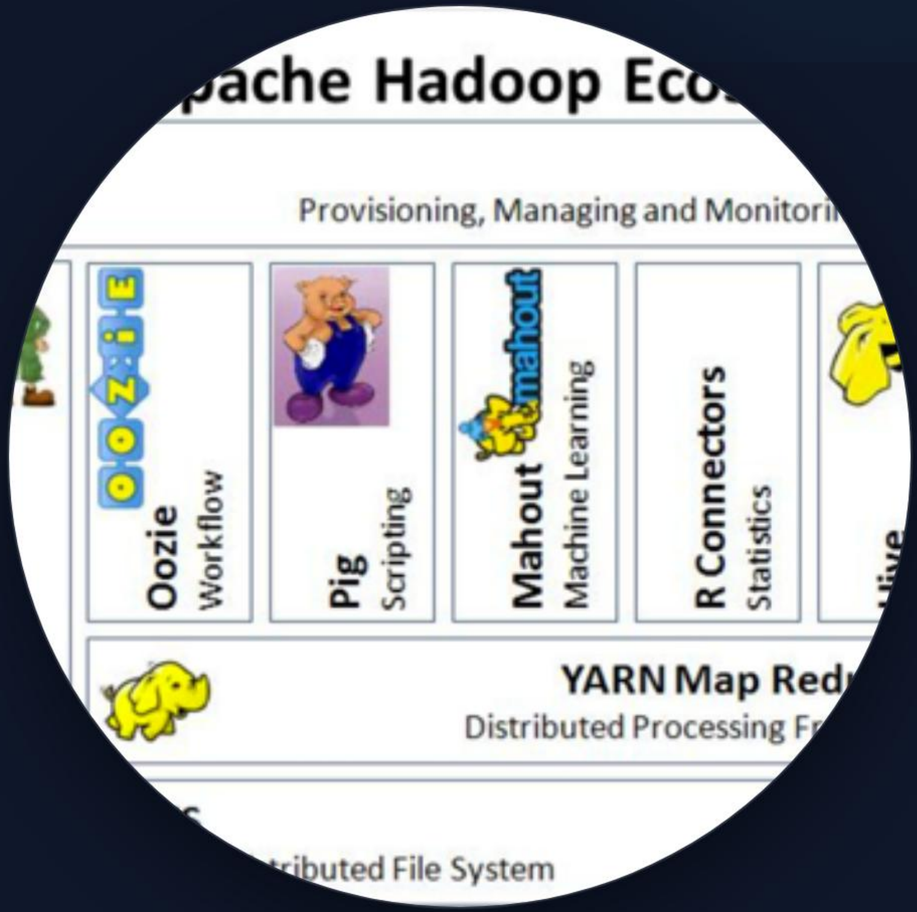
"Oozie manages the dependency and timing of complex multi-stage data pipelines."

9. KAFKA: STREAMING BASICS

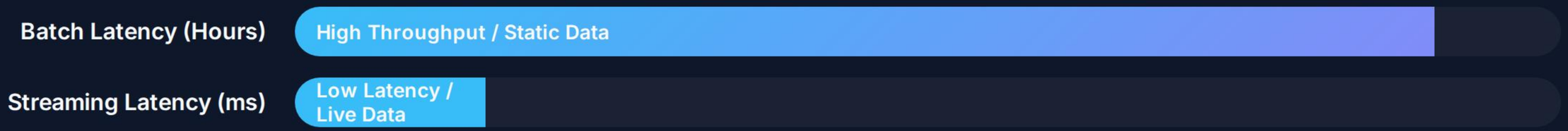
The Message Backbone

Kafka is a distributed **event store** and stream-processing platform.

- **Producers:** Send data to topics.
- **Topics:** Categories used to organize messages.
- **Consumers:** Read data from topics.



10. REAL-TIME VS. BATCH



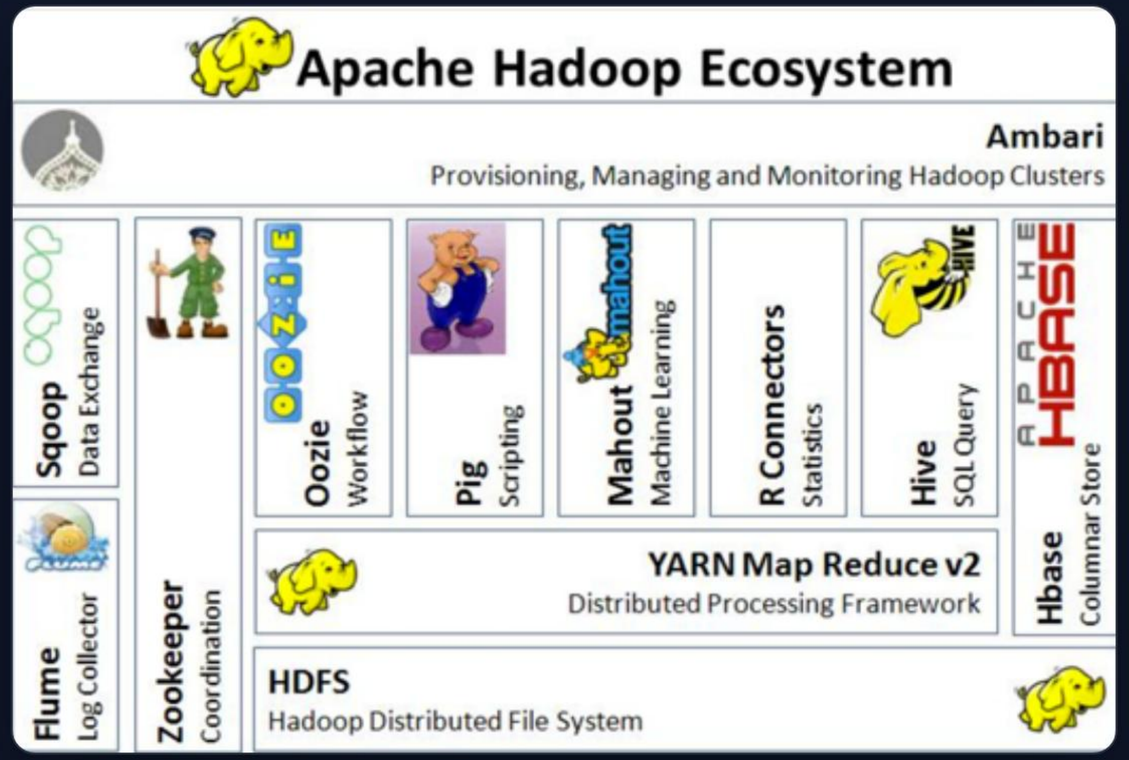
Batch

Data collected over time, processed in chunks. Best for payroll, billing.

Real-time

Data processed immediately upon arrival. Best for fraud, IoT.

11. REAL-TIME PIPELINES



Streaming Architectures

Architectures like **Lambda** or **Kappa** are used to handle live feeds.

Kafka acts as the buffer, and Spark Streaming/Flink acts as the processing engine to deliver sub-second insights.

12. BATCH JOBS OVERVIEW

Standard Daily Workload:

- Extraction of full DB snapshots via Sqoop.
- Heavy transformation (Join/Sort) in Spark Core.
- Writing back to Data Warehouse (Hive) for BI.

13. STREAMING CONCEPTS

Windowing

Processing data in temporal buckets (e.g., last 5 mins).

Watermarking

Handling late-arriving data in the stream.

Checkpointing

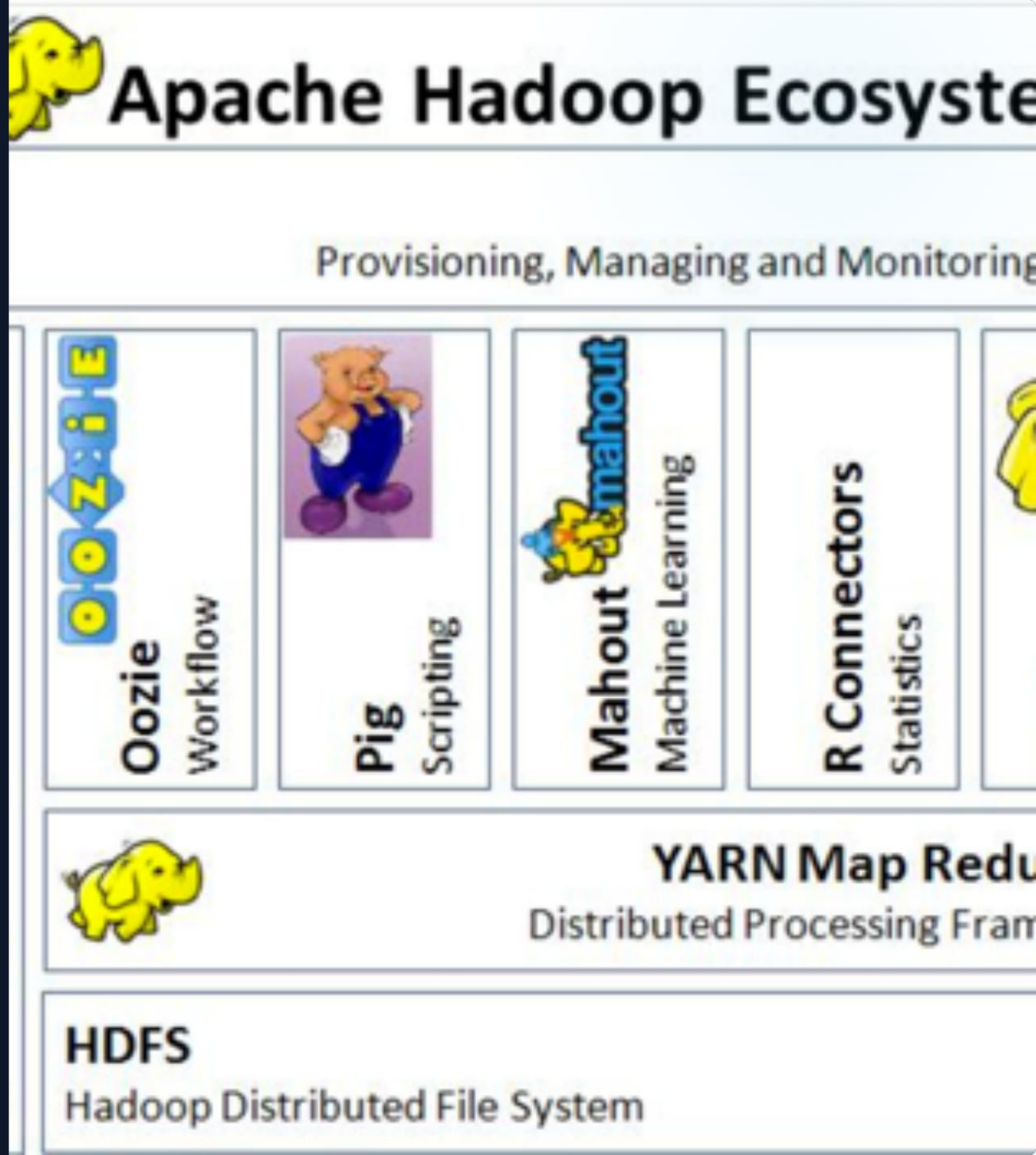
Ensuring exactly-once processing through state recovery.

14. CASE: BANKING

Fraud Detection

Processing millions of credit card transactions per second. Real-time pipelines identify anomalies (velocity/location) to block fraudulent attempts instantly.

Tools: Kafka → Spark Streaming → HBase.

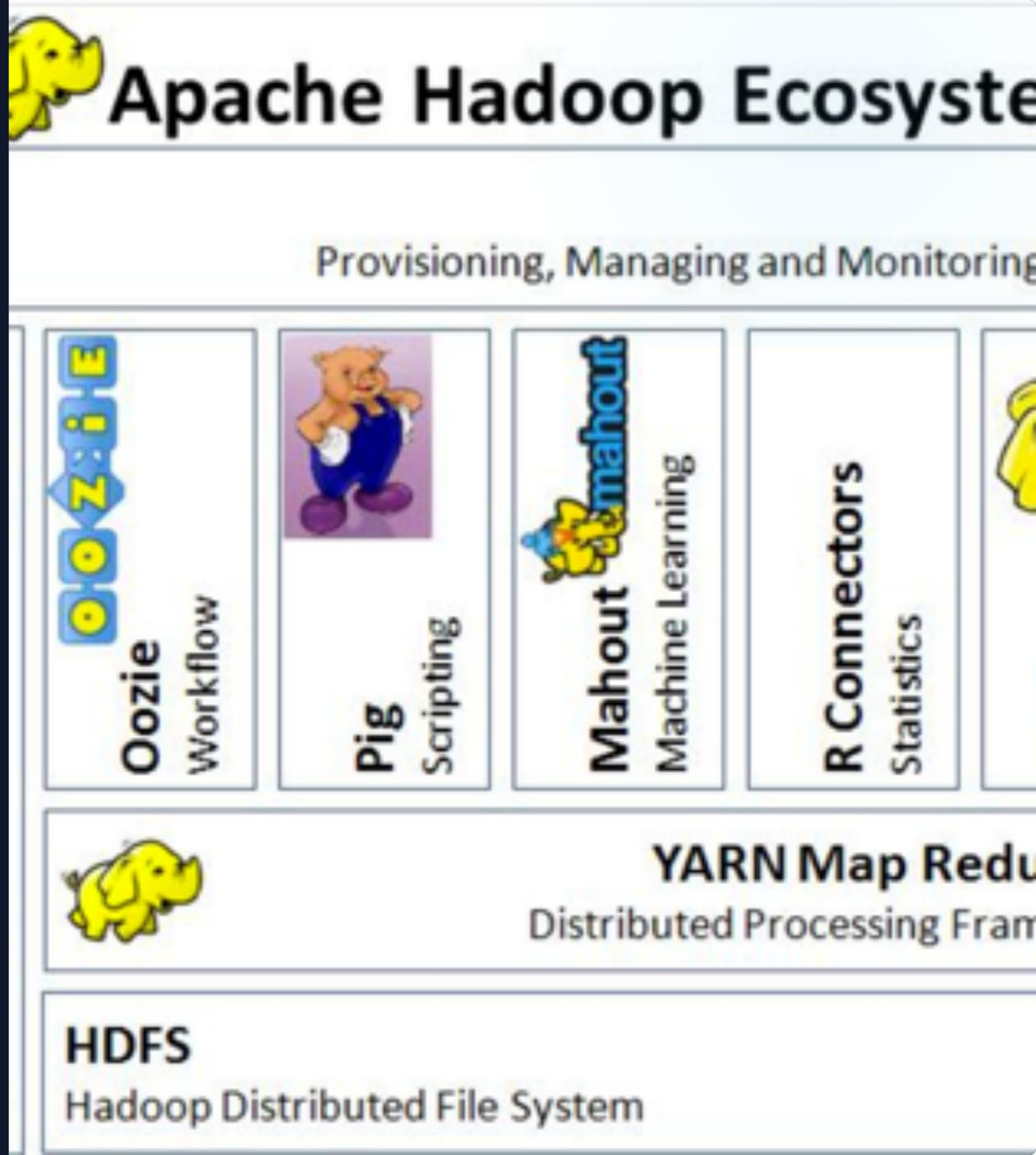


15. CASE: HEALTHCARE

Predictive Diagnostics

Aggregating patient EMR data with live vital monitoring to predict critical events like sepsis or heart failure before they occur.

Tools: Flume → HDFS → Spark MLlib.

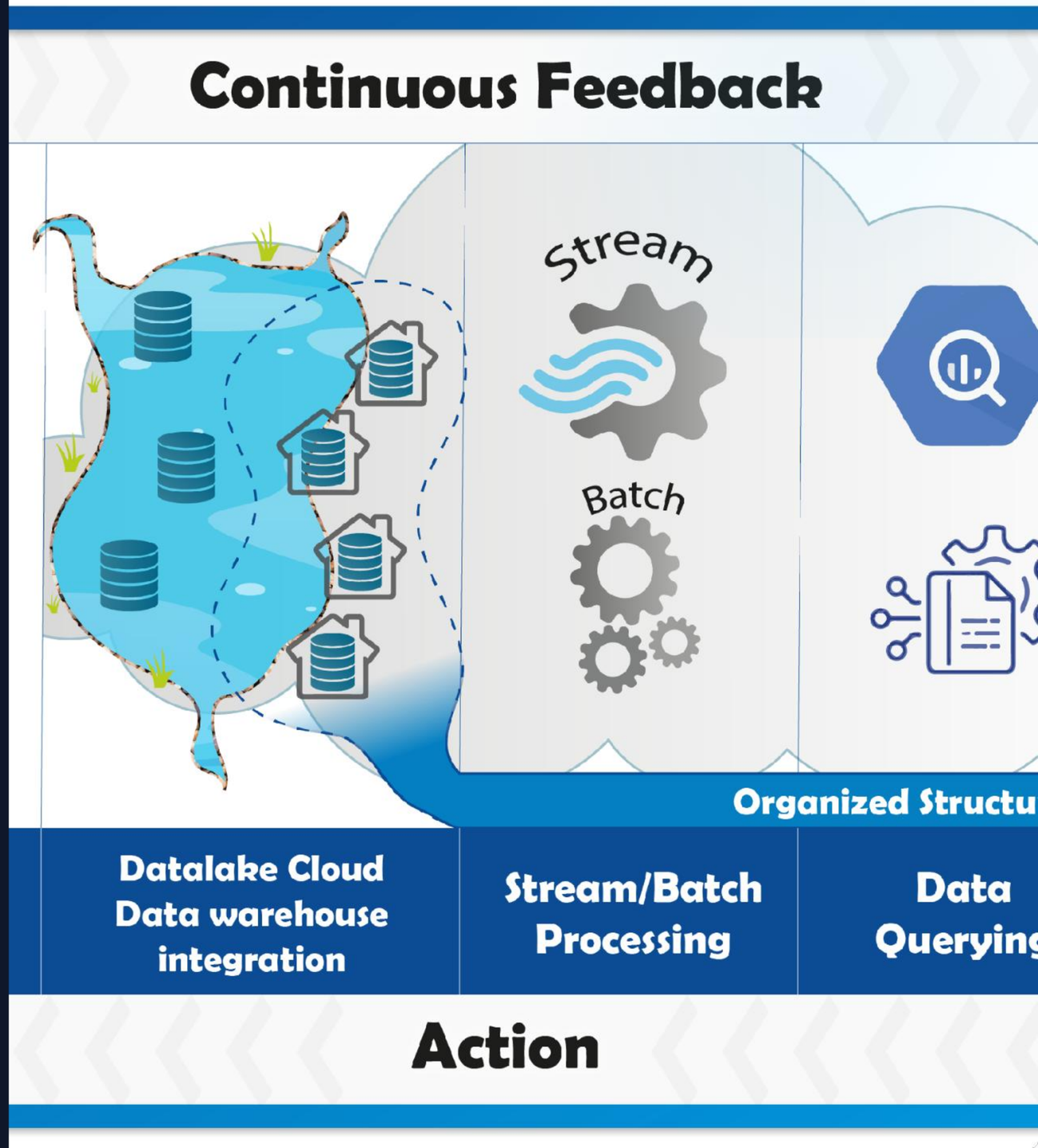


16. CASE: E-COMMERCE

Personalization

Capturing clickstream data to update product recommendations while the user is still on the site, increasing conversion rates.

Tools: Kafka → Spark Structured Streaming.

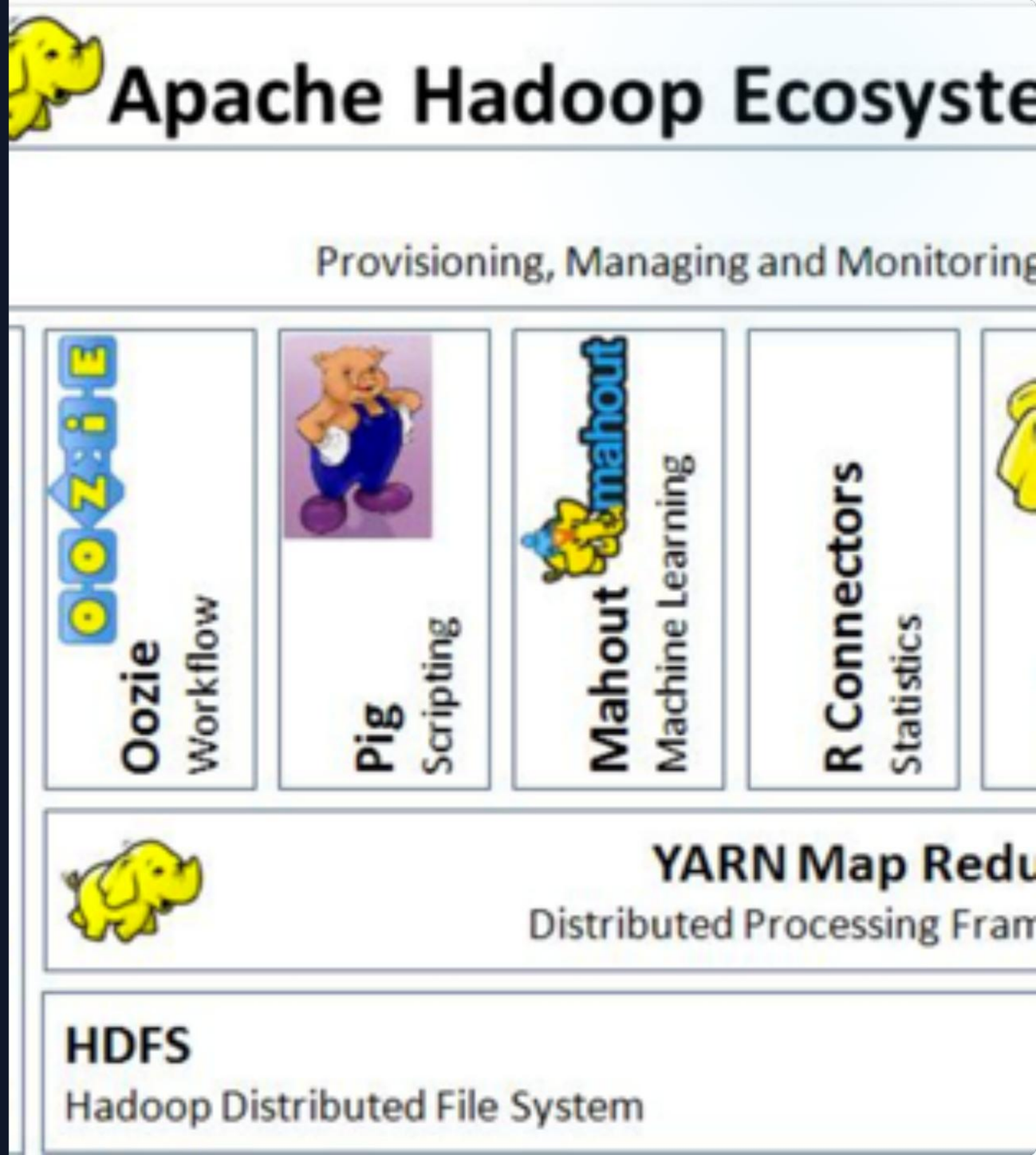


17. CASE: IOT & LOGS

Fleet Monitoring

Analyzing sensor data from thousands of vehicles to optimize routes and schedule preventive maintenance, reducing downtime.

Tools: MQTT → Kafka → Spark → TimeSeries DB.



18. HANDS-ON ROADMAP

1. Data Import

Use Sqoop to ingest relational data into HDFS.

2. Log Stream

Set up Flume to stream web logs into HDFS.

3. Spark Job

Run a WordCount or Join job on the Hadoop cluster.

```
$ spark-submit --master yarn --deploy-mode cluster my_script.py
```

Q & A

Advanced Distributed Computing Session

admin@bigdata-hub.edu cluster-computing.tech

IMAGE SOURCES



https://miro.medium.com/1*_k-1zXSuH4oFv70lo2GIlA.png

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